

Umakant Kulkarni

Contact

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Current Address

New Providence, NJ 07974 USA

RESEARCH INTERESTS

Cellular Networks, Distributed Systems, Network Security, Resource Management, Orchestration, Autoscaling, Wi-Fi Networks, Objective QoE, Reinforcement Learning, Transformers, LLMs

EDUCATION

Ph.D., Computer Science

Purdue University, West Lafayette, USA

GPA 4.0

Jan 2021 - Dec 2025

Master of Science, Computer Systems Networking

Northeastern University, Boston, USA

GPA 3.71

Sep 2015 - May 2017

Bachelor of Engineering, Electronics and Telecommunication

University of Pune, India

First Class with Distinction

Aug 2011 - May 2015

PROFESSIONAL EXPERIENCE

Nokia Bell Labs, Murray Hill, NJ USA

Research Scientist

Jan 2026 – Present

- Conducting research at the intersection of cellular networks (5G/6G), distributed systems, network security, and AI/ML to advance next-generation mobile network systems.

Affirmed Networks/Microsoft, Acton MA USA

Principal Software Engineer

Sep 2018 – Dec 2020

- Developed stateless, containerized telecommunication software based on 5G standards for core network functions AMF, SMF, UPF, N3IWF, PCF and UDM that can be deployed via docker and/or kubernetes to achieve scalability, resiliency and redundancy.
- Implemented Istio service-mesh features into 5G NFs for observability, security and traffic management.
- Designed a build environment to define, install and upgrade applications on kubernetes using Helm.
- Developed DPDK containerized UE-gNB simulator for testing different data-plane functionality.
- Built pyATS-enabled automation infrastructure and integrated it with JIRA and Jenkins for simulating CI/CD model of cloud-native microservice architecture.

Software Quality Assurance Engineer

Jun 2017 – Aug 2018

- Contributed towards architectural development and quality assurance for Virtualized ePDG (evolved Packet Data Gateway), 'Security-Gateway' and T-WAG (Trusted - Wireless Access Gateway) network functions from 4G Evolved Packet Core.
- Wrote test plans and automated call-flows related to AAA, PCRF and OFCS Servers, GTP-Path Management, X.509 Security Certificates over S2x, SWm, STa and SWu interfaces, in Tcl and Python.
- Performed system, regression, functional, performance and redundancy testing on Affirmed Network's Mobile Cloud Content using manual and automation methods.

RESEARCH EXPERIENCE

Purdue University, West Lafayette, IN USA

Jan 2021 – Dec 2025

- Graduate Research Assistant in Prof. Sonia Fahmy's research group
- Worked on 6G-focused transformers/LLMs for efficient network orchestration

Indian Institute of Science, Education and Research, Pune India

May 2013 – Jul 2013

- Contributed towards development of Atomic Force Microscopy for Pico level measurements to study various aspects of molecular mobility, dynamics and mechanical response at single molecular scale.

TEACHING EXPERIENCE

Purdue University, West Lafayette, IN USA

Jan 2021 – Dec 2021

- Graduate Teaching Assistant for the course 'Computer Architecture (CS 250)'

Northeastern University, Boston, MA USA

Jan 2017 – May 2017

- Graduate Teaching Assistant for the course 'Telecommunication Architecture and Systems (TELE 5320)' for 2 sections.

INTERNSHIPS

InterDigital, Conshohocken PA USA

Research Intern

May 2025 – Aug 2025

- Analyzed, designed, and implemented next generation 6G mobile networks.
- Developed 6G networking protocols, services, and solutions to enable application verticals (GenAI).

Hewlett Packard Enterprise, Milpitas CA USA

Research Associate Intern

May 2022 – Dec 2023

- Characterized QoE of volumetric video streaming over enterprise Wi-Fi networks, analyzing the impact of Wi-Fi control parameters and comparing QoE with traditional 2D video applications.
- Proposed an adaptive method for Wi-Fi resource allocation to maximize the overall system QoE and QoE fairness.

Affirmed Networks, Inc. Acton MA USA

May 2016 – Jan 2017

- Laid the foundation for ePDG and T-WAG automation infrastructure; Built the 'smoke' as well as 'sanity' frameworks to automate the end-to-end workflow.
- Modified open-source libraries and configured StrongSwan module onto a linux box as well as Raspberry Pi to support different Wi-Fi calling features over IPsec.

Thuse Electronics Pvt. Ltd., Pune India

Nov 2013 – Dec 2013

- Worked on the processes transformer manufacturing, PCB design, PCB assembling in line production, quality and rework management, process quality control, wave soldering on RoHS and non RoHS sections, automatic optical inspection and surface mount technology.

CERTIFICATIONS

Certified Telecommunication Engineer (Reg. No. – S13PUN128)
by Bharat Sanchar Nigam Limited (BSNL), Govt. of India

- Platinum Level - Mobile Communication, IP Networking and Cyber Security Jul 2014
- Gold Level - Broadband Technology, Optical Fiber Technology Jan 2014
- Silver Level - Digital Switching and Transmission System, Telecom Support Infrastructure Jul 2013

PUBLICATIONS

- Janus: A Dual-Mask Attention Transformer for Log-based Anomaly Detection in Cellular Networks
Umakant Kulkarni, Sonia Fahmy
Proc. ACM Meas. Anal. Comput. Syst. 10, 1, Article 14, (SIGMETRICS'26), 36pp., March 2026.
- RUN: A Case for Cross-Layer Networked Virtual Reality
Yufeng Chen, **Umakant Kulkarni**, Voicu Popescu, Sonia Fahmy
Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), 10 pp., October 2025.
- Maestro: QoE-Aware Dynamic Resource Allocation in Wi-Fi Networks
Umakant Kulkarni, Khaled Diab, Lianjie Cao, Faraz Ahmed, Shivang Aggarwal, Puneet Sharma, Sonia Fahmy
Proceedings of the ACM on Networking, Volume 3, Issue CoNEXT1, Article 4, 24 pp., March 2025
- Securing the Cloud-Native 5G Control Plane
Umakant Kulkarni, Sonia Fahmy
IEEE MILCOM Workshop on Secure FutureG Wireless Communications and Networked Systems, 6 pp., October 2024
- Understanding the Impact of Wi-Fi Configuration on Volumetric Video Streaming Applications
Umakant Kulkarni, Khaled Diab, Shivang Aggarwal, Lianjie Cao, Faraz Ahmed, Puneet Sharma, Sonia Fahmy
ACM SIGCOMM Workshop on Emerging Multimedia Systems (EMS), 6 pp., September 2023
- Toward QoE-based Routing Path Selection
Umakant Kulkarni, Yufeng Chen, Patrick Melampy, Sonia Fahmy
IEEE International Conference on High Performance Switching and Routing (HPSR), 6 pp., June 2023
- Slicure5G: Secure Slicing for 5G (Poster)
Umakant Kulkarni and Sonia Fahmy
USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2 pp. April 2023
- Towards A Low-Cost Stateless 5G Core (Poster)
Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN), 2 pp., July 2022
- The Cost of Stateless Network Functions in 5G
Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS), 7 pp., December 2021

TECHNICAL SKILLS

- Languages: C/C++, Golang, Python, Java, Matlab, JavaScript | Scripting: Bash, Tcl/Tk, Perl
- OS: Windows, Linux, UNIX, iOS
- Tools/Technologies: Kubernetes, Docker, Helm, Istio, Fluent-d, Prometheus, Grafana, Kiali, MongoDB, Jaeger, Kibana, AWS, Terraform
- ML: Keras, TensorFlow, PyTorch, CUDA, NumPy, Scikit-Learn, Pandas

PROFESSIONAL SERVICE

- *USENIX NDSI 2026* — Artifact Evaluation Committee
- *ACM SIGCOMM 2025* — Artifact Evaluation Committee
- *ACM IMC 2025* — Shadow Technical Program Committee
- **Ad hoc journal reviews:**
 - *IEEE Internet Computing*
 - *IEEE Transactions on Networking*

FELLOWSHIPS AND AWARDS

- Elected to membership in The Honor Society of Phi Kappa Phi in January 2025. Membership is by invitation only and recognizes academic excellence among the top 10% of graduate students.
- Received employee merit recognition award under the “High-Profile Projects” category for achieving significant milestones in “Securing 5G for Military Operations” research project at Purdue University
- 3rd Prize in Best-in-class 2022 Summer Interns Project Fair organized by Hewlett Packard Enterprise
- Received “Outstanding Graduate Teaching Assistant Award” for significant accomplishments and highest ratings in teaching and mentoring students at College of Engineering, Northeastern University.
- Awarded by KVPY Fellowship (English: Young Scientist Incentive Plan) with All India Rank – 5. This research fellowship was given by The IISc, Bangalore and Department of Science and Technology.
- Received Best Paper/Research/Poster award at seven regional conferences, India (2011-2015).

WEB PROFILES

Homepage: <https://umakantkulkarni.github.io>

Google Scholar: <https://scholar.google.com/citations?hl=en&user=1v3d0A0AAAAJ>

LinkedIn: <https://www.linkedin.com/in/umakantkulkarni/>

GitHub: <https://github.com/UmakantKulkarni/>

REFERENCES

- Prof. Sonia Fahmy
Professor of Computer Science, Purdue University
Email: fahmy@purdue.edu
- Mark Libby
System Software Manager, Boston Dynamics
Email: mark@mwlibby.net
- Prof. Peter O'Reilly
Director, Telecommunication Networks program, Northeastern University
Email: p.oreilly@northeastern.edu